Vyakyarth vs LaBSE — Baseline & Fine-tuning Comparison

Evaluated on IN22-Conv | All 22 Indic Languages | 462 Directed Pairs |

1. Overview

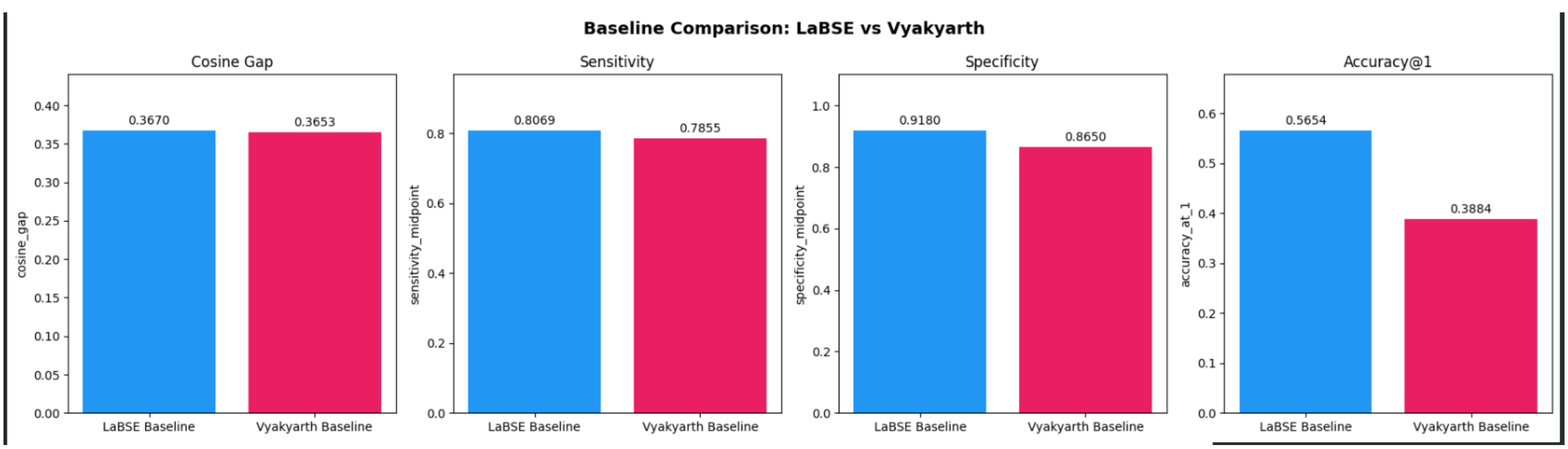
This report compares Vyakyarth (krutrim-ai-labs/Vyakyarth) against LaBSE on the IN22-Conv benchmark, covering both base models and their fine-tuned variants. Vyakyarth was fine-tuned using the same strategy and hyperparameters as LaBSE: single top-pair models, single bottom-pair models, combined top-5, combined bottom-5, and a mixed top+bottom model. A total of 25 models were evaluated (2 baselines + 13 LaBSE fine-tuned + 23 Vyakyarth fine-tuned), all on the same unseen IN22-Conv dataset (1,503 sentences, 462 directed Indic-Indic pairs, all 22 scheduled languages).

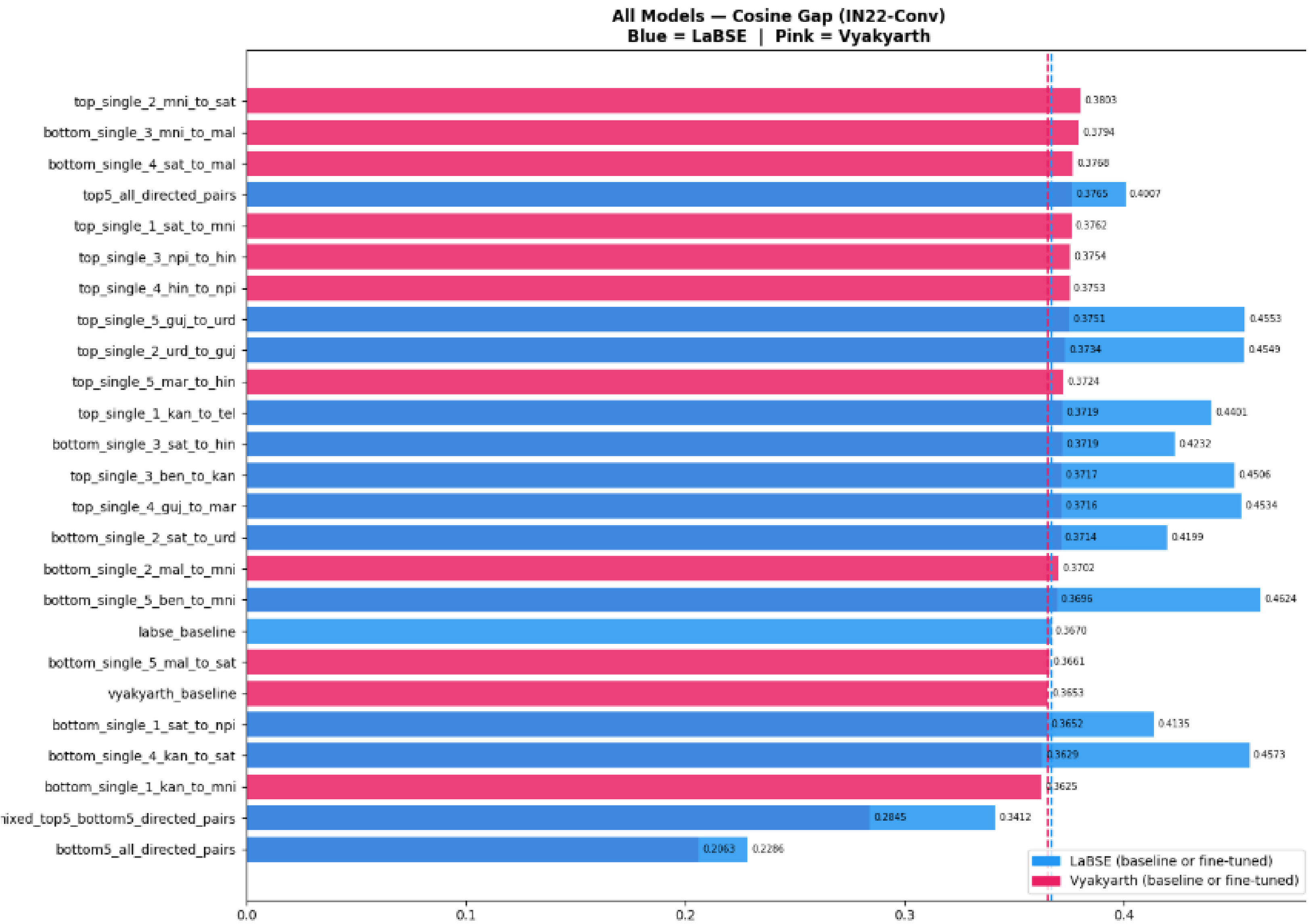
2. Baseline Comparison — Vyakyarth vs LaBSE

Before any fine-tuning, here is how the two base models compare directly:

Model	Cosine Gap	Sensitivity	Specificity	Accuracy@1
LaBSE baseline	**0.3670**	**0.8069**	**0.9180**	**0.5654**
Vyakyarth baseline	**0.3653**	**0.7855**	**0.8650**	**0.3884**

Cosine gap is nearly identical between the two baselines (0.3670 vs 0.3653). However, LaBSE shows clearly stronger specificity (0.9180 vs 0.8650) and a much higher accuracy@1 (0.5654 vs 0.3884) — a 17.7 percentage point gap. This means Vyakyarth, while showing similar raw discrimination at the aggregate level, is meaningfully worse at the practical retrieval task of finding the correct translation among 22 candidates.





3. Best Fine-tuned Model per Strategy

For each fine-tuning strategy, the best-performing model from each base model is shown below.

Strategy	Best Model	Cosine Gap	Sensitivity	Specificity
Single Top (LaBSE)	top_single_5_guj_to_urd	0.4553	0.8336	0.9257
Single Top (Vyakyarth)	top_single_2_mni_to_sat	0.3803	0.7905	0.8805

Single Bottom (LaBSE)	bottom_single_5_ben_to_mni	0.4624	0.8867	0.9320
Single Bottom (Vyakyarth)	bottom_single_3_mni_to_mal	0.3794	0.7878	0.8812
Combined Top5 (LaBSE)	top5_all_directed_pairs	0.4007	0.8192	0.9272
Combined Top5 (Vyakyarth)	top5_all_directed_pairs	0.3765	0.7978	0.8556
Combined Bottom5 (LaBSE)	bottom5_all_directed_pairs	0.2286	0.8893	0.7750
Combined Bottom5 (Vyakyarth)	bottom5_all_directed_pairs	0.2063	0.8746	0.5995
Mixed (LaBSE)	mixed_top5_bottom5	0.3412	0.8929	0.8592
Mixed (Vyakyarth)	mixed_top5_bottom5	0.2845	0.8798	0.7416

4. Key Observations

4.1 LaBSE outperforms Vyakyarth at every strategy

Across every single fine-tuning strategy — single top, single bottom, combined top5, combined bottom5, and mixed — the LaBSE-based model beats the corresponding Vyakyarth-based model on cosine gap, sensitivity, specificity.

4.2 Vyakyarth degrades more severely under combined training

The Vyakyarth combined-bottom5 model collapses to accuracy@1 of 0.1651 specificity of 0.5995 — barely better than random guessing. The corresponding LaBSE model, while also weak (accuracy@1 0.3945), degrades far less severely. This suggests Vyakyarth's embedding space is more fragile under multi-pair training interference.

4.3 Both models follow the same fine-tuning pattern

Despite the performance gap, the qualitative pattern is identical for both base models: single-pair fine-tuning improves over baseline, combined/mixed training causes regression, and bottom-pair fine-tuning generalises at least as well as top-pair fine-tuning. This consistency suggests the pattern is a property of the fine-tuning strategy itself, not specific to either base model.

4.4 Vyakyarth's best fine-tuned model still underperforms LaBSE baseline